2025-08-11 - Analyze the data and perform interval testing for sampling distributions.

11 August 2025 16:35

→ A small business recieves on average 12 customers per day. What is the probability that the business will recieve exactly 8 customers in one day?

$\lambda = 12$; $x = 8$

$$P(x=8) = \frac{\lambda^{x} \cdot e^{-\lambda}}{x!} = \frac{12^{8} \cdot e^{-12}}{8!}$$

$$= 0.0655 \approx 6.55\%$$

⇒ A student recieves on average 7 text messages in a 2 hour period. What is the probability that the student will recieve exactly 9 text messages in a 2 hour period?

$\lambda = 7$; $x = 9$

$$P(x=9) = \frac{\lambda^{x} \cdot e^{-\lambda}}{x!} = \frac{7^{9} \cdot e^{-7}}{9!} = 0.1014$$

$$\approx 10.14\%$$

↳ What is the probability that the student will recieve exactly 24 messages in an will receive exactly 24 messages in the 8 hour period?

$$\times 4 \left(\begin{array}{ll} \lambda = 7 & t = 2 \\ \lambda = 28 & t = 8 \end{array} \right) \times 4$$

$x = 24$

$$\therefore P(x=24) = \frac{28^{24} \cdot e^{-28}}{24!} = 0.060\ldots$$

$$\approx \underline{\underline{6.01\%}}$$

⇒ A small business receives on average 8 calls per hour. What is the probability that the business will receive exactly 7 calls in an hour?

$\lambda = 8 \; ; \quad x = 7$

$$P(x=7) = \frac{8^7 \cdot e^{-8}}{7!} = \underline{\underline{13.96\%}}$$

↳ What is the probability that the business will receive at most 5 calls in one hour?

$\lambda = 8 \; ;$

$$P(x \leq 5) = P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4) + P(x=5)$$

$$e^{-8} \left[\frac{8^0}{0!} + \frac{8^1}{1!} + \frac{8^2}{2!} + \frac{8^3}{3!} + \frac{8^4}{4!} + \frac{8^5}{5!} \right]$$

$$e^{-8}\left[\frac{8^0}{0!}+\frac{8^1}{1!}+\frac{8^2}{2!}+\frac{8^3}{3!}+\frac{8^4}{4!}+\frac{8^5}{5!}\right]$$

$$= 0.1912 \approx \underline{19.12\%}$$

→ What is the probability that the business will recieve more than 6 calls in 1 hour?

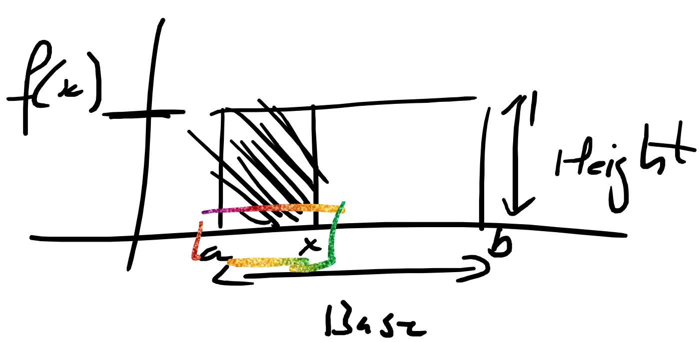
$\lambda = 8$; $P(x>6)$

$$P(x>6) = 1 - P(x\le 6)$$

$$= 1 - e^{-8}\left[\frac{8^0}{0!}+\frac{8^1}{1!}+\frac{8^2}{2!}+\frac{8^3}{3!}+\frac{8^4}{4!}+\frac{8^5}{5!}+\frac{8^6}{6!}\right]$$

$$= 0.6866 \approx \underline{68.7\%}$$

→ Uniform Distribution :



$$\text{Area} = \text{Base} \times \text{Height}$$

$$= (x-a) \times f(x) \approx \boxed{f(x) = \frac{1}{b-a}}$$

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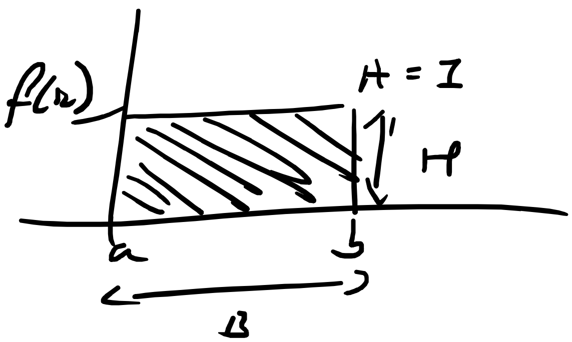
$$\boxed{= (x-a) \cdot \left(\frac{1}{b-a}\right)}$$

⇒ The amount of time a person must wait for a train to arrive in a certain town is uniformly distributed b/w 0 & 40 minutes. Determine the PDF $f(x)$.

$A = 1$; $f(x)$



$A = \text{Base} \times \text{Height}$

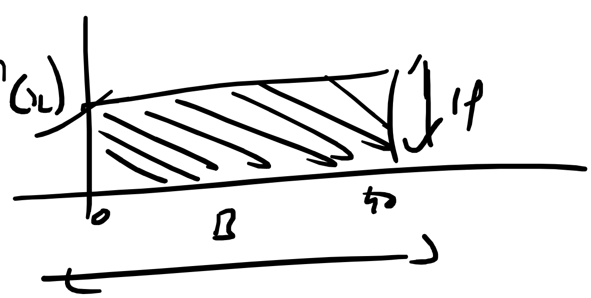
$1 = (b-a) \cdot f(x)$

$\therefore f(x) = \frac{1}{b-a}$

$\therefore f(x) \times (b-a) = \left(\frac{1}{\cancel{b-a}}\right) \times \cancel{(b-a)}$

$\therefore A = (b-a)\, f(x)$

; $f(x)$



$a = 0$; $b = 40$

$$f(x) = \frac{1}{b-a} = \frac{1}{40-0}$$

$$= \boxed{1/40}$$

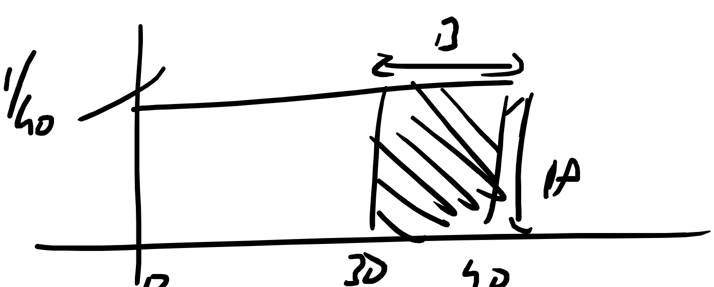
↳ What is the probability that a person must wait less than 8 minutes?

$P(x<8) = ?$

$$A = B \cdot H = 8 \times \left(\frac{1}{40}\right)$$

$$= 8/40 = 1/5 = \boxed{20\%}$$

↳ What is the probability that a person must wait more than 30 min?



$P(x>30) = P(30<x<40)$

$A = B \cdot H$

$P(X > 30)$

$$A = B \cdot H$$

$$= (40-30) \times \left(\frac{1}{40}\right) = \frac{10}{40} = \frac{1}{4} = \boxed{25\%}$$